asterics — A Tool for the ExploRation and Integration of omiCS data

Élise Maigné[1,*], Céline Noirot[1,2,*], Jérôme Mariette[1,2], Yaa Adu Kesewaah[1,3], Sébastien Déjean[3,4], Camille Guilmineau[1,3], Julien Henry[1,3], Arielle Krebs[1,2], Laurence Liaubet[5], Fanny Mathevet[1,3], Hyphen-Stat[6], Christine Gaspin[1,2], Nathalie Vialaneix[1,3]

[1] Université de Toulouse, INRAE, UR MIAT, 31326, Castanet-Tolosan, France – [2] Université Fédérale de Toulouse, INRAE, BioinfOmics, GenoToul Bioinformatics facility, 31326, Castanet-Tolosan, France – [3] Plateforme Biostatistique, Genotoul, Toulouse, France – [4] IMT, UMR5219, Université de Toulouse, CNRS, UPS, 31062, Toulouse, France – [5] GenPhySE, Université de Toulouse, INRAE, ENVT, F-31326, Castanet-Tolosan France – [6] Hyphen-stat, https://hyphen-stat.com/, Toulouse, France
(*) These authors contributed equally to the work.

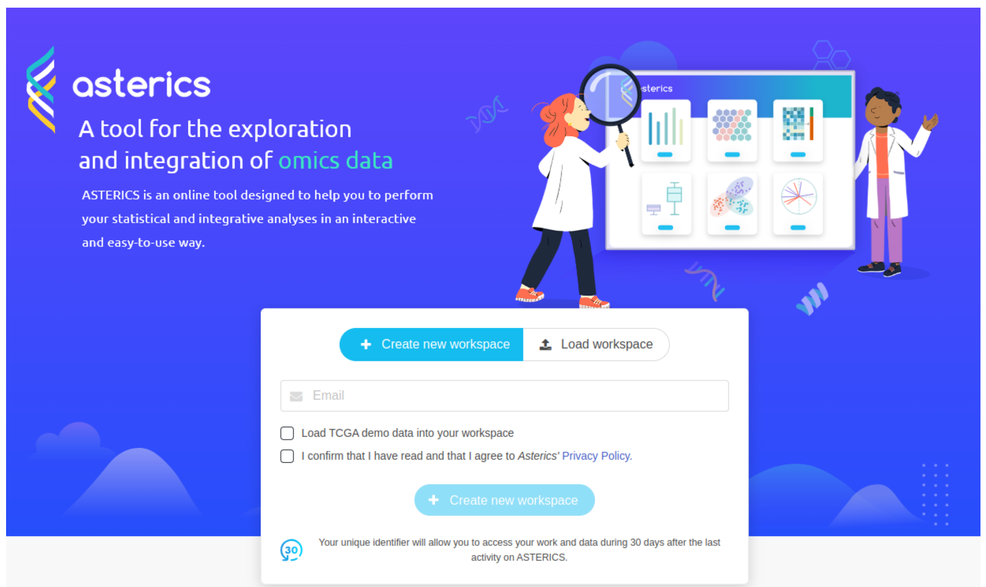
ASTERICS web application

Explore, integrate and visualize your omics data.

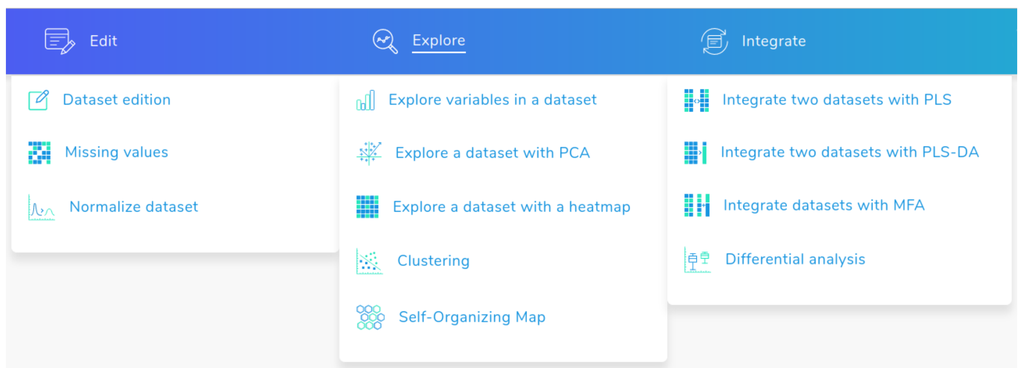
backend → **frontend**

Rserve {renv} ⟷ PyRServe / Flask ⟷ Vue.js / BULMA

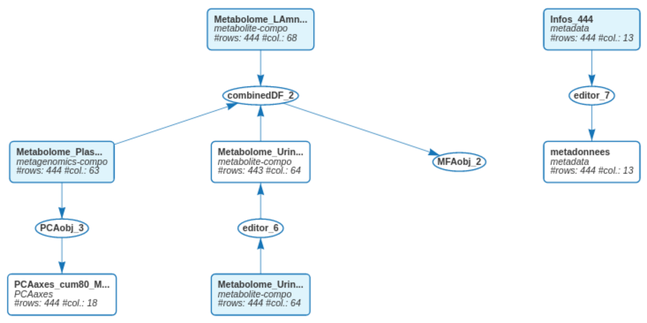
- Available at: http://asterics.miat.inrae.fr
- Source code (GPLv2): https://forgemia.inra.fr/asterics/asterics



Available analyses

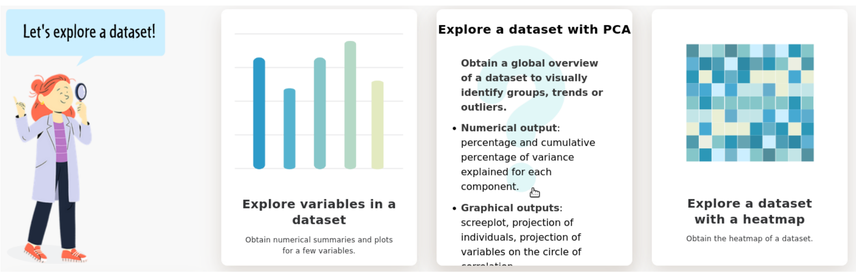


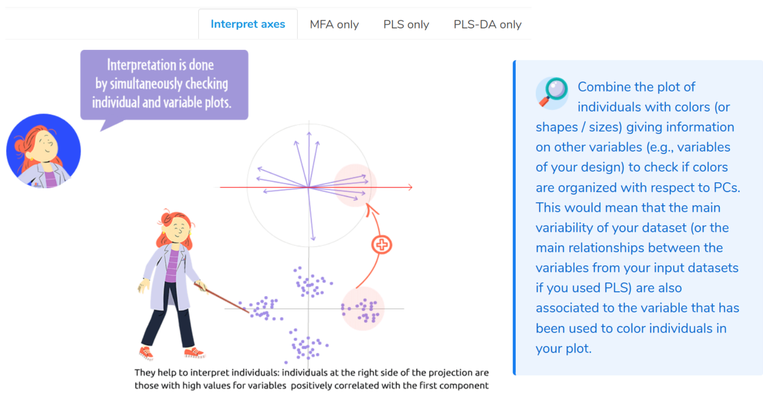
Generated datasets and analyses are navigable using a **DAG**:



Help pages

On each analysis page, **focused help pages** are available to help the user choose analysis settings or interpret the results. These pages include images and GIF animations.

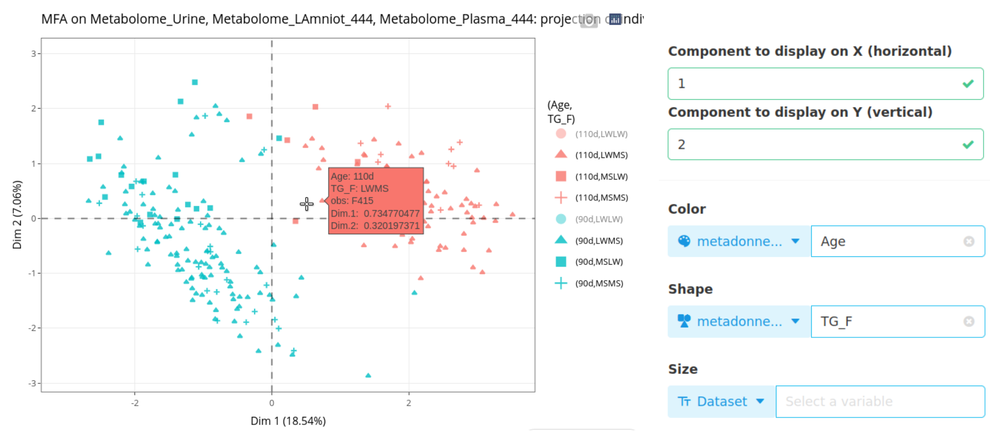




Full documentation:
https://asterics.pages.mia.inra.fr/user_documentation/

Outputs of analyses

- Interactive and easily customizable plots:



- Sortable and filtrable tables:

Post-hoc test results for protein vs clinical. (Wilcoxon tests, corrected with 'BH')

Variables	Conditions	p-values	Adj. p-values	Signs
	left lower			
14.3.3_epsilon	left lower inner quadrant-left	0.6071	1	-1
14.3.3_epsilon	left lower outer quadrant-left	0.5617	1	-1
14.3.3_epsilon	left lower outer quadrant-left lower inner quadrant	0.9426	1	1
14.3.3_epsilon	left upper inner quadrant-left lower inner quadrant	0.563	1	1

Additional features

- Use your **UUID** to come back later to your analyses.
- Export the results of your analysis in **HTML reports** (using Rmarkdown).
- **Export** complex objects as RDS data or `data.frame` as CSV files.
- Use the online tool or use `Docker` for a local use or for deployment.

Contributors

Partners

GenoToul Biostat · Hyphen from data to science · Institut de Mathématiques de Toulouse · GenoToul Bioinfo · INRAE

Funding

La Région Occitanie Pyrénées · Méditerranée